



Foundations of Programming

Assignment No: 8

PROBLEM STATEMENT:

Write a program in **C** to accept marks of five courses of a student and compute the result. A student is considered **PASS** if he/she scores **40 marks or more in each course**. If the student passes, calculate the aggregate percentage and assign the grade as follows:

- Aggregate $\geq 75\%$: **Distinction**
- Aggregate $\geq 60\%$ and $< 75\%$: **First Division**
- Aggregate $\geq 50\%$ and $< 60\%$: **Second Division**
- Aggregate $\geq 40\%$ and $< 50\%$: **Third Division**

OBJECTIVE:

- To understand the use of **conditional statements** in C
- To apply **logical operators** for decision making
- To calculate **total, percentage, and grade** of a student
- To enhance problem-solving skills using real-life scenarios

THEORY:

In academic evaluation systems, student performance is assessed based on individual subject marks as well as overall aggregate performance. In C programming, such evaluation systems can be implemented using conditional statements like **if**, **else if**, and **else**. Logical operators help in checking multiple conditions simultaneously. Percentage calculation involves arithmetic operations, and grading logic is implemented based on predefined ranges. This program demonstrates structured programming and decision-making in C.

ALGORITHM:

1. Start
2. Input marks of five subjects
3. Check if any subject mark is less than 40:
4. If yes $\rightarrow$ Display FAIL and stop

5. Calculate total marks = sum of all five subjects
6. Calculate percentage = total / 5
7. Determine grade:
 - If percentage $\geq 75 \rightarrow$ Distinction
 - Else if percentage $\geq 60 \rightarrow$ First Division
 - Else if percentage $\geq 50 \rightarrow$ Second Division
 - Else if percentage $\geq 40 \rightarrow$ Third Division
8. Display total marks, percentage, result (PASS), and grade
9. Stop

PLATFORM: Linux

PROGRAM

```
1  #include <stdio.h>
2
3  int main() {
4      int m1, m2, m3, m4, m5;
5      int total;
6      float percentage;
7
8      printf("Enter marks of five subjects: ");
9      scanf("%d %d %d %d %d", &m1, &m2, &m3, &m4, &m5);
10
11     // Check pass condition
12     if (m1 < 40 || m2 < 40 || m3 < 40 || m4 < 40 || m5 < 40) {
13         printf("\nResult: FAIL\n");
14     } else {
15         total = m1 + m2 + m3 + m4 + m5;
16         percentage = total / 5.0;
17
18         printf("\nTotal Marks = %d", total);
19         printf("\nPercentage = %.2f%", percentage);
20         printf("\nResult: PASS");
21
22         // Grade calculation
23         if (percentage >= 75)
24             printf("\nGrade: Distinction");
25
26         else if (percentage >= 60)
27             printf("\nGrade: First Division");
28         else if (percentage >= 50)
29             printf("\nGrade: Second Division");
30         else if (percentage >= 40)
31             printf("\nGrade: Third Division");
32     }
33     return 0;
34 }
```

INPUT:

```
Enter marks of five subjects: 40 50 75 95 100
```

OUTPUT: The program displays the total marks, percentage, result, and grade.

```
Total Marks = 360
Percentage = 72.00%
Result: PASS
Grade: First Division

=== Code Execution Successful ===
```

CONCLUSION: Thus, the program to calculate student result and grade based on marks was successfully implemented using C programming. This program demonstrates the effective use of conditional statements, logical operators, and arithmetic operations.

FAQs:

1. What is the objective of this C program?
2. Why is an if-else ladder used in this program?
3. Which data types are generally used in this program?

Answers:-

1. The objective is to calculate a student's total marks, percentage, result, and grade using conditional statements and logical operators.
2. An if-else ladder is used to check multiple conditions (grade ranges) and execute the correct block based on the percentage.
3. `int` - for marks and total
`float` - for percentage